

M. SIVA NAGA RAJU

Senior Technical Analyst | AI Solutions Architect | Agentic Migration Platform Specialist

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Portfolio: <https://sivanagarajumolabant.github.io/Portfolio/>

SUMMARY

Senior Technical Analyst with **7+ years of experience** at Quadrant Technologies, specializing in enterprise-grade AI-powered database migration and release automation for the QMigrator product. Integrated and productionized a multi-agent migration ecosystem across Release Agent, Migration Agent, API Agent, UI Agent, DB Agent, DevOps Agent, Testing Automation, and execution triage/remediation flows, using LangGraph, FastAPI, Azure DevOps, and Google's A2A/ADK-style agent-to-agent SDK patterns.

- Integrated independently developed agent services into a unified QMigrator release workflow with A2A communication, status tracking, DevOps automation, and test feedback loops
- Achieved 98% automation in database schema conversion across Oracle, PostgreSQL, SQL Server, Snowflake, Databricks, Sybase, and SQL Server modernization paths
- Reduced migration error resolution time by 75% — from 2 hours to 8 minutes — through AI triage, remediation, validation, and rerun loops
- Delivered enterprise migration automation for UHG, Elevance Health/Carelon, eBay, Nike, and Apollo Hospitals

SKILLS

AI & Generative AI: LangChain, LangGraph, Multi-Agent Systems, Google A2A/ADK-style Agent SDK Integration, RAG Architecture, Prompt Engineering, Agentic AI, Pydantic AI, NLP

Agent Orchestration: FastAPI Agent Services, A2A JSON-RPC, Agent Cards, Webhook Signals, LangGraph Checkpointing, Human-in-the-Loop Review, Multi-Agent Release DAGs

Large Language Models: Azure OpenAI GPT-4, Anthropic Claude, Google Gemini, Groq, Ollama

Vector Databases: ChromaDB, FAISS, Pinecone, Milvus, Qdrant

AI Evaluation: DeepEval, Ragas, AI Guardrails, AI Safety, Quality Assurance

Embeddings: OpenAI Embeddings, HuggingFace Embeddings, Ollama Embeddings

Programming Languages: Python (Expert), SQL (Advanced), JavaScript

Backend Frameworks: FastAPI, Django, Django REST Framework, Flask, Streamlit

API Development: RESTful APIs, A2A Protocol APIs, JSON-RPC, GraphQL, Swagger, OpenAPI

Data Engineering: Pandas, NumPy, Apache Airflow, Celery, ETL Pipelines, Matplotlib, Seaborn

Databases: PostgreSQL, Oracle, SQL Server, MySQL, MongoDB, Redis

Data Platforms: Snowflake, Databricks, Delta Lake

Database Skills: Query Optimization, Performance Tuning, PL/SQL, PL/pgSQL, T-SQL, Index Design

Cloud - Azure: AKS, ACR, Azure DevOps, Key Vault, Pipelines, Azure PostgreSQL, Azure Blob Storage

Cloud - AWS: ECS, Fargate, ECR, API Gateway, Lambda, S3, EC2, RDS

DevOps & Containers: Docker, Kubernetes, Helm Charts, ArgoCD, Azure Pipelines, GitHub Actions, GitLab CI/CD

Frontend: Angular 17, ReactJS, JavaScript, TypeScript, HTML5, CSS3, Material UI

Testing: Playwright Automation, Pytest, Selenium, E2E Status Polling, Automated Remediation Loops

Tools: Git, GitHub, GitLab, Bitbucket, JIRA, Confluence, Postman, VS Code, PyCharm, Cursor, Windsurf

WORK EXPERIENCE

Quadrant Technologies — Hyderabad, India

Senior Technical Analyst | March 2019 - Present (7+ Years)

Product: QMigrator — Enterprise Database Migration Platform

Core Product Developer and AI Solutions Architect | 2019 - Present

Platform URL: <https://qcam.qmigrator.ai/>

QMigrator is Quadrant Technologies' flagship enterprise database migration platform that automates end-to-end schema conversion, data migration, and code translation across heterogeneous database environments. Key contributor since inception — built and evolved the platform from a feature management system to a full-scale AI-powered migration solution.

- Built and maintained the core migration platform serving 15+ enterprise clients over 7 years
- Evolved the product from manual feature-based conversion (2019) to fully autonomous AI-driven migration (2025)
- Processed 100,000+ database objects across Oracle, PostgreSQL, SQL Server, Snowflake, and Databricks
- Delivered 15+ large-scale enterprise migrations with 98% automation rate

QMigrator Development: Agentic Release and Migration Automation Platform | 2025 - Present

Integrated and productionized a multi-agent QMigrator automation platform. Several component agents were developed by different contributors/teams; my role focused on end-to-end integration, orchestration, request/status lifecycle, A2A communication, DevOps hand-off, testing automation, and production readiness.

Release Agent — Multi-Agent Orchestration and Status Control

- Integrated Release Agent as the central controller for migration releases, using a configurable agent DAG to run Assessment, Conversion, Data Migration, Testing Workload, API, UI, DB, and DevOps components
- Implemented Google A2A/ADK-style agent-to-agent integration using Agent Cards, JSON-RPC message/send, task polling, webhook signals, and shared job correlation
- Added release request persistence, component-level status tracking, checkpointer-backed recovery, DevOps build status polling, and QCam UI progress updates
- Stack: Python, FastAPI, LangGraph, PostgreSQL checkpointing, Google A2A SDK patterns, Azure DevOps, YAML configuration

Migration Agent — AI Migration Path Generation Engine

- Integrated Assessment, Conversion, Data Migration, and Testing Workload agents as standalone LangGraph workflows exposed through REST and A2A server endpoints
- Supported migration path generation, reference selection, template adaptation, validation/fix loops, PR creation, artifact publishing, and upstream hand-off between agents
- Extended the platform to allow common dispatcher/main file updates when required while preserving existing function signatures
- Stack: Python, FastAPI, LangGraph, LangChain-style tool loops, Azure OpenAI/Claude/Gemini/Ollama, Azure Blob, Azure DevOps PR APIs

API/UI/DB Agent Integration — Product Surface Automation

- Integrated API Agent for QMigrator backend assessment controller scaffolding, .NET operation libraries, route constants, DI registration, and database-specific operations
- Integrated UI Agent for Angular 17 migration-type wiring across constants, services, enums, connection screens, schema extraction, E2E migration, and configuration flows
- Integrated DB Agent for database repository branch and pull-request automation as part of the release component flow
- Stack: FastAPI, Angular 17, .NET/QMigrator API conventions, Azure DevOps, Git automation

DevOps Agent and Testing Automation — Deployment Validation Loop

- Integrated DevOps Agent for Azure DevOps pipeline triggering, ArgoCD bootstrap, Azure project bootstrap, SaaS infra/Terraform setup, release summaries, and build remediation support
- Added pipeline build status polling with simple status outputs: running, success, failed
- Integrated Testing Automation after DevOps build success, including external trigger/status APIs, 60-second polling, E2E result interpretation, AI triage/remediation, redeploy, and retry loops
- Stack: Azure DevOps, ArgoCD, Kubernetes, Docker, Terraform workflows, Playwright/E2E automation APIs, FastAPI

QMigrator Development: Multi-LLM Infrastructure | 2025 - Present

- Architected enterprise-grade LLM abstraction layer supporting Azure OpenAI, Anthropic Claude, Groq, Google Gemini, and Ollama

- Built dynamic model switching with cost optimization, fallback mechanisms, and token usage tracking
- Integrated AI guardrails and evaluation frameworks (DeepEval, Ragas) for quality assurance and safety
- Standardized agent communication surfaces with A2A/ADK-style contracts so independently built agents could be discovered, invoked, tracked, and chained consistently
- Stack: Python, LangChain, DeepEval, Ragas, JSON/YAML Configuration

QMigrator Development: Core Migration Engine | 2021 - 2022

- Developed core migration engine components with modular Python architecture
- Built Django REST Framework APIs and React-based UI for migration workflow management
- Designed extensible plugin architecture supporting 50+ migration feature types

QMigrator Development: QBook V2 — Feature Management Platform | 2020 - 2021

- Created self-service platform enabling developers to register and manage migration features
- Built multi-level migration types, approval workflows, and version control for feature modules
- Developed full-stack Django + React application with admin dashboard serving 30+ developers

QMigrator Development: QBook V1 — Feature Management System | 2019 - 2020

- Established foundational architecture for feature-based migration approach
- Implemented developer registration, authentication, and approval workflows
- Built Python-based feature templates for standardized development across teams

QMigrator Client Deployments

Project: CARELON HEALTH / ELEVANCE HEALTH — Sybase to SQL Server Migration

Ongoing Migration Automation Support | 2026 - Present

- Supporting ongoing Sybase to SQL Server migration work under the Carelton Health / Elevance Health account
- Applying the integrated QMigrator agent ecosystem for migration-path generation, API/UI/DB integration, DevOps orchestration, and Testing Automation validation
- Coordinating release flow across Migration Agent, Release Agent, DevOps Agent, QCam UI, and execution triage/remediation components
- Focused on reducing manual release coordination by standardizing request tracking, pipeline status, E2E testing feedback, and remediation loops

Project: NIKE — Cloud Data Platform Modernization

Snowflake to Databricks Migration | 2026 - Present

- Leading large-scale data platform migration for enterprise analytics on Databricks
- Designed ETL pipeline transformations achieving 98% data migration accuracy with automated validation
- Optimized query performance with 40% faster execution times on Databricks
- Deployed on AWS infrastructure with Delta Lake architecture for data lakehouse implementation

Project: eBay — Enterprise Database Migration

Oracle to PostgreSQL Migration | 2024 - 2026

- Deployed Stage 1 and Stage 2 conversion agents for automated error correction pipeline
- Implemented Performance Agent for query optimization during migration
- Achieved 98% automation in schema conversion with AI-driven validation
- Deployed on AWS using ECS with Fargate, ECR, and API Gateway

Project: ELEVANCE HEALTH (Formerly Anthem) — Enterprise Migration

SQL Server to Azure PostgreSQL | 75,000+ Objects | 2024 - 2025

- Orchestrated AI agent workflows for complex PL/SQL to PL/pgSQL conversions
- Implemented Debug Mode for human-in-the-loop validation of critical business logic

- Achieved 95% conversion accuracy for stored procedures and functions
- Reduced manual intervention by 80% through intelligent agent automation

Project: UHG (United Health Group) — Enterprise Migration

Oracle to PostgreSQL | 50,000+ Objects, 2TB+ Data | 2023 - 2025

- Deployed all 6 AI agents to production serving 100+ concurrent users
- Achieved 98% schema conversion automation with AI-driven error correction
- Implemented Kubernetes-based architecture for horizontal scaling
- Reduced migration timeline by 60% compared to traditional approaches with zero-downtime deployment

Project: APOLLO HOSPITALS — Healthcare System Migration

On-Premise Oracle to Azure PostgreSQL | 2022 - 2024

- Managed weekly release cycles with comprehensive deployment validations
- Coordinated cross-functional teams (DBA, Application, Infrastructure) for seamless rollouts
- Ensured 100% compliance with healthcare data security standards (HIPAA)
- Achieved zero data loss during mission-critical healthcare system migration

EDUCATION

Master of Technology (M.Tech) in Computer Science and Engineering

KITS Engineering College, JNTU Kakinada — Kakinada, Andhra Pradesh

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Chalapathi Engineering College, JNTU Kakinada — Guntur, Andhra Pradesh

Diploma in Electrical and Electronics Engineering

AI-Huda Polytechnic College, AICTE — Nellore, Andhra Pradesh

CERTIFICATIONS

- Advanced LangChain and LangGraph Patterns for Production Systems
- Azure Solutions Architect — Cloud Infrastructure and DevOps
- Kubernetes Application Development and Container Orchestration
- Database Performance Tuning and Query Optimization
- Agile and Scrum Methodologies — Sprint Planning and Team Leadership

LANGUAGES

- English: Professional Working Proficiency
- Telugu: Native
- Hindi: Professional Working Proficiency